Gustavo Pérez

Machine Learning Researcher | Computer Vision and Statistical ML for Reliable Image-Based Measurement

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About me: My research develops machine learning and computer vision methods for large-scale image analysis under uncertainty, with an emphasis on statistical estimation and human-in-the-loop systems. I study how to obtain reliable global measurements from imperfect models, with applications in climate science, ecology, and scientific imaging.

Education

Doctorate of Philosophy in Computer Science **(Fulbright Scholar)** — 2018 – 2024
University of Massachusetts, Amherst
Advisor: Subhransu Maji

Master of Science in Biomedical Engineering — 2016 – 2018
Universidad de los Andes, Colombia
Advisor: Pablo Arbeláez

Bachelor of Science in Electronic Engineering — 2004 – 2008
Universidad del Norte, Colombia

Research Experience

Postdoctoral Scholar (BAIR Lab) — 2024 – 2026
University of California, Berkeley (EECS)
Advisors: Stella Yu & Miki Lustig

Visiting Scholar — 2024 – 2026
University of Michigan, Ann Arbor (CSE)

Graduate Research Assistant (Computer Vision Lab) — 2019 – 2024
University of Massachusetts, Amherst (CICS)

Graduate Research Assistant (Biomedical Computer Vision Lab) — 2018 – 2018
Universidad de los Andes, Colombia

Scholarships & Awards

Best Paper Award for the AI for Social Impact track at AAAI 2024, Vancouver, Canada — 2024
Best Paper Award 1st Runner-Up at BuildSys 2023, Istanbul, Turkey — 2023
CICS Outstanding Synthesis Project Award. UMass Amherst, MA — 2021
1st place at the ISBI 2018 Lung Nodule Malignancy Prediction Challenge. ISBI 2018, Washington DC — 2018
Fulbright scholarship. Colciencias-Fulbright Cohort 2018, Bogotá, Colombia — 2017
- Funding awarded : \$330.000.000 COP ∼\$110.000 USD

Best project of the faculty award. EEII 2016, Universidad de los Andes, Bogota, Colombia — 2016

Publications

Refereed Conference Proceedings

1. **Normalize Filters ! Classical Wisdom for Deep Vision**
   Gustavo Perez, Stella Yu
   Advances in Neural Information Processing Systems (NeurIPS), 2025
2. **Human-in-the-Loop Visual Re-ID for Population Size Estimation**
   Gustavo Perez, Daniel Sheldon, Grant Van Horn, Subhransu Maji
   The European Conference on Computer Vision (ECCV), 2024
3. **DISCount : Counting in Large Image Collections with Detector-Based Importance Sampling**
   Gustavo Perez, Subhransu Maji†, Daniel Sheldon†
   Association for the Advancement of Artificial Intelligence (AAAI), 2024 (†equal advising)
   Best Paper Award for the AI for Social Impact Track

4. Taking the long view : Enhancing learning on multi-temporal, highres, and disparate remote sensing data
   Santiago Correa, Gustavo Perez, Paulina Jaramillo, Jay Taneja
   The 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (ACM BuildSys), 2023
   Best Paper Award 1st Runner-Up
5. Domain Adaptors for Hyperspectral Images
   Gustavo Perez, Subhransu Maji
   26TH International Conference on Pattern Recognition (ICPR), 2022
6. Automated Detection of Lung Nodules with Three-dimensional Convolutional Neural Networks
   Gustavo Perez, Pablo Arbelaez
   Proc. SPIE 10572, 13th International Conference on Medical Information Processing and Analysis, 2017 (oral)

Journal Articles

7. Aggregating three sources of long-term trends of swallows and martins to identify priority conservation areas in the Great Lakes region
   Maria Belotti, Brian Gerber, Wenlong Zhao, Yuting Deng, Victoria Simons, Gustavo Perez, Jeffrey Kelly, Subhransu Maji, Daniel Sheldon, Kyle Horton
   Journal of Applied Ecology, 2025
8. Pairing weather radars with eBird data for large-scale monitoring of swallow and martin communal roosts
   Maria Belotti, Wenlong Zhao, Yuting Deng, Victoria Simons, Gustavo Perez, Jeff Kelly, Subhransu Maji, Dan Sheldon, Kyle Horton
   Ornithological Applications, 2025
9. Representation Learning for Long-Chain Hydrocarbon Adsorption in Zeolites
   Yachan Liu, Ping Yang, Gustavo Perez, Aaron Sun, Wei Fan, Subhransu Maji, Peng Bai
   Journal of Materials Chemistry A, 2025
10. Continental Connections : Changing Temperature, Wind and Precipitation Advance the Postbreeding Roosting Phenology of Avian Aerial Insectivores
    Yuting Deng, Birgen Haest, Maria Belotti, Wenlong Zhao, Gustavo Perez, Elske Tielens, Dan Sheldon, Subhransu Maji, Jeffrey Kelly, Kyle Horton
    Global Ecology and Biogeography, 2025
11. Feedback in emerging extragalactic star clusters, FEAST : The relation between 3.3 μm PAH emission and Star Formation Rate traced by ionized gas in NGC 628
    Benjamin Gregg, Daniela Calzetti, Angela Adamo, Varun Bajaj, Jenna E. Ryon, Sean T. Linden, Matteo Correnti, Michele Cignoni, Matteo Messa, Elena Sabbi, John S. Gallagher, Kathryn Grasha, Alex Pedrini, Robert A. Gutermuth, Jens Melinder, Ralf Kotulla, Gustavo Perez, Mark R. Krumholz, Arjan Bik, Goran Ostlin, Kelsey E. Johnson, Giacomo Bortolini, Linda J. Smith, Monica Tosi, Subhransu Maji, Helena Faustino Vieira
    The Astrophysical Journal (ApJ), 2024
12. Using spatio-temporal information in weather radar data to detect and track communal roosts
    Gustavo Perez†, Wenlong Zhao†, Zezhou Cheng, Maria Carolina Belotti, Yuting Deng, Victoria Simons, Elske Tielens, Jeffrey Kelly, Kyle Horton, Subhransu Maji Subhransu Maji, Daniel Sheldon
    Remote Sensing in Ecology and Conservation, 2024 (†equal contribution)
13. ZeoNet : 3D convolutional neural networks for predicting adsorption in nanoporous zeolites
    Yachan Liu†, Gustavo Perez†, Zezhou Cheng, Aaron Sun, Samuel Hoover, Wei Fan, Subhransu Maji, Peng Bai
    Journal of Materials Chemistry A, 2023 (†equal contribution)
14. Long-term analysis of persistence and size of swallow and martin roosts in the US Great Lakes
    Maria Belotti, Yuting Deng, Wenlong Zhao, Victoria Simons, Zezhou Cheng, Gustavo Perez, Elske Tielens, Subhransu Maji, Daniel Sheldon, Jeffrey Kelly, and Kyle Horton
    Remote Sensing in Ecology and Conservation, 2023
15. Quantifying long-term phenological patterns of aerial insectivores roosting in the Great Lakes region using weather surveillance radar
    Yuting Deng, Maria Belotti, Wenlong Zhao, Zezhou Cheng, Gustavo Perez, Elske Tielens, Victoria Simons, Daniel Sheldon, Subhransu Maji, Jeffrey Kelly, Kyle Horton
    Global Change Biology, 2023

16. **Star Cluster Formation and Evolution in M101 : An Investigation with the Legacy Extragalactic UV Survey**
Sean Linden, Gustavo Perez, Daniela Calzetti, Subhransu Maji Subhransu Maji, Matteo Messa, Bradley Whitmore, Rupali Chandar, Angela Adamo, Kathryn Grasha, David Cook, Bruce Elmegreen, Daniel Dale, Elena Sacchi, Elena Sabbi, Eva Grebel, Linda Smith
The Astrophysical Journal (ApJ), 2022

17. **Lung Nodule Malignancy Prediction in Sequential CT Scans : Summary of ISBI 2018 Challenge**
Yoganand Balagurunathan, Andrew Beers, Michael McNitt-Gray, Lubomir Hadjiiski, Sandy Napel, Dmitry Goldgof, Gustavo Perez, Pablo Arbelaez, Alireza Mehrtash, Tina Kapur, Ehwa Yang, Jung Won Moon, Gabriel Bernardino Perez, Ricard Delgado-Gonzalo, M. Mehdi Farhangi, Amir A. Amini, Renkun Ni, Xue Feng, Aditya Bagari, Kiran Vaidhya, Benjamin Veasey, Wiem Safta, Hichem Frigui, Joseph Enguehard, Ali Gholipour, Laura Castillo, Laura Daza, Paul Pinsky, Jayashree Kalpathy-Cramer, Keyvan Farahani
IEEE Transactions on Medical Imaging, 2021

18. **StarcNet : Machine Learning for Star Cluster Identification**
Gustavo Perez, Matteo Messa, Daniela Calzetti, Subhransu Maji, Dooseok Jung, Angela Adamo, Mattia Sirressi
The Astrophysical Journal (ApJ), 2021

19. **Automated lung cancer diagnosis using three-dimensional convolutional neural networks**
Gustavo Perez, Pablo Arbelaez
Medical & Biological Engineering & Computing, 2020

Preprints & Ongoing Work

20. **Sequence-Agnostic Motion Correction for Brain MRI with MIMO RF Motion Sensing**
Gustavo Perez, Suma Anand, Stella Yu, Miki Lustig
(Ongoing work), 2025

21. **Nationwide Building-Attribute Benchmarking and Scalable Zero-Shot Extraction with Foundation Models**
Gustavo Perez†, Zilin Wang†, Brian Wang, Fei Pan, Frank McKenna, Stella Yu
Under submission, 2025 (†equal contribution)

Refereed Workshop & Short Papers

22. **Learning Non-Rigid Motion From MIMO RF Navigators**
Rinni Bhansali, Thomas Olausson, Gustavo Perez, Philippa Krahn, Alessandro Sbrizzi, Michael Lustig
Proceedings of the Annual Meeting of ISMRM, 2026 (to appear)

23. **Representation Learning for Predicting Shape Selectivity in Nanoporous Zeolites**
Yachan Liu, Ping Yang, Aaron Sun, Zezhou Cheng, Gustavo Perez, Wei Fan, Subhransu Maji, Peng Bai
The 29th North American Catalysis Society Meeting, NAM29, 2025

24. **Representation Learning for All-Silica Zeolites : Model Representations, Transfer Learning, and Multi-Task Learning.** Yachan Liu, Ping Yang, Gustavo Perez, Aaron Sun, Wei Fan, Subhransu Maji, Peng Bai
AIChE Annual Meeting, AIChE, 2024

25. **Developing 3D Convolutional Neural Networks (ZeoNet) for Predicting Adsorption in Nanoporous Zeolites**
Yachan Liu, Gustavo Perez, Zezhou Cheng, Aaron Sun, Sam Hoover, Wei Fan, Subhransu Maji, Peng Bai
AIChE Annual Meeting, AIChE, 2023

26. **A Semi-Automated System to Annotate Communal Roosts in Large-Scale Weather Radar Data**
Wenlong Zhao, Gustavo Perez, Zezhou Cheng, Maria Belotti, Yuting Deng, Victoria Simons, Elske Tielens, Jeffrey Kelly, Kyle Horton, Subhransu Maji, Daniel Sheldon
Computational Sustainability Workshop, NeurIPS, 2023

27. **An AI-Assisted Labeling Tool for Cataloging High-Resolution Images of Galaxies**
Gustavo Perez, Sean Linden, Timothy McQuaid, Matteo Messa, Daniela Calzetti, Subhransu Maji
AI for Science : Progress and Promises Workshop, NeurIPS, 2022

28. **Finding Four-Leaf Clovers : A Benchmark for Fine-Grained Object Localization**
Laura Bravo†, Alejandro Pardo†, Gustavo Perez†, Pablo Arbelaez
Fine-Grained Visual Categorization Workshop (FGVC6), CVPR, 2019 (†equal contribution)

29. **Lung Nodules Detection in CT images using Computer Vision**
Gustavo Perez
VIII International Seminar in Biomedical Engineering. ISSN 2322-7702. Universidad de los Andes, Colombia, 2016

Academic Service

Area chair

- Neural Information Processing Systems–**NeurIPS** (2026–to serve)

Honors Thesis (499Y/T) committee member

- Student : Advait Gosai, University of Massachusetts, Amherst (2024)
  Thesis : *"Leveraging Vision-Language Models for Adaptive Query-Based Image Counting"*

Reviewer for the following international conferences

- IEEE/CVF Conference on Computer Vision and Pattern Recognition–**CVPR** (2025)
- International Conference on Learning Representations–**ICLR** (2025)
- International Conference on Computer Vision–**ICCV** (2023, 2025)
- European Conference on Computer Vision–**ECCV** (2024)
- Asian Conference on Computer Vision–**ACCV** (2024)
- Association for the Advancement of Artificial Intelligence–**AAAI** AI for Social Impact Track (2022–2026)
- IEEE/CVF Winter Conference on Applications of Computer Vision–**WACV** (2023–2025)
- European Association for Signal Processing–**EUSIPCO** (2019–2024)
- Symposium of Image, Signal Processing, and Artificial Vision–**STSIVA** (2019, 2021)
- International Conference on Image, Video Processing and Artificial Intelligence–**IVPAI** (2018)

Reviewer for the following scientific journals

- International Journal of Computer Vision–IJCV (2023–2024)
- IEEE Transactions on Geoscience and Remote Sensing–IEEE TGRS (2023–2024)
- Scientific Reports (2023)
- BMC Medical Informatics and Decision Making (2024)
- Annals of Translational Medicine (2020)
- Oral Diseases (2020)
- Medical & Biological Engineering & Computing (2019, 2024)
- Revista TecnoLógicas (2024)
- IEEE Transactions on Cybernetics (2018)

Reviewer for the following workshops at international conferences

- ICLR Workshop on Weight Space Learning–**WSL** (2025)
- Computational Sustainability workshop–**CompSust** (2023)
- Fine-Grained Visual Categorization workshop–**FGVC** (2021–2024)
- LatinX in AI/CV workshop–**LXAI/LXCV** (2021–2024)

Mentorship & Advising

Mentored PhD, MSc, and undergraduate students across computer vision, scientific ML, and interdisciplinary applications.

Mentored students

- Zilin Wang. **Scalable Zero-Shot Building-Attribute Extraction.** *Ph.D. student*. University of Michigan, 2025
- Max Hamilton. **Machine Learning for Star Cluster Classification.** *Ph.D. student*. UMass Amherst, 2024-25
- Santiago Correa. **Deep Learning from Multi-Temporal Satellite Data.** *Ph.D. student*. UMass Amherst, 2021-23
- Dooseok Jung. **Machine Learning for Star Cluster Classification.** *Ph.D. student (Astronomy Dept.)*. UMass Amherst, 2019-21
- Laura Bravo. **Benchmarking Fine-Grained Object Localization.** *M.Sc. student*. Universidad de los Andes, 2017-18
- Alejandro Pardo. **Benchmarking Fine-Grained Object Localization.** *M.Sc. student*. Universidad de los Andes, 2017-18

UBC Master of Data Science Mentoring Program (2025-26) website

- **Student :** Wendy Frankel. *Master's student*. University of British Columbia

Early Research Scholar Program mentor* (2023-24) website

*ERSP is a dual-mentored research apprenticeship for undergraduates to address the underrepresentation of minority students in computing

- **Project :** Using Computer Vision for Counting Things
- **Students :**
  Mantra Burugu. *Undergraduate student*. University of Massachusetts, Amherst
  Lucky Kovvuri. *Undergraduate student*. University of Massachusetts, Amherst
  Tan Le. *Undergraduate student*. University of Massachusetts, Amherst
  Larissa Zhu. *Undergraduate student*. University of Massachusetts, Amherst

Teaching Experience

Guest Lecturer 2026
University of Michigan, Ann Arbor
Course: Advanced Topics in Computer Vision (*Topic:* From Data to Science with AI and Human-in-the-Loop)

Guest Lecturer 2025
Universidad Autónoma de Occidente, Colombia
Course: Medical Imaging Using AI (*Topic:* Introduction to Deep Learning and Applications in Biomedical Imaging)

Guest Lecturer 2023
University of Massachusetts, Amherst
UMass Machine Learning Club (*Topic:* Counting in Large Image Collections with Detector-Based Importance Sampling)

Graduate Teaching Assistant 2018 – 2019
University of Massachusetts, Amherst
Course: Computer Systems Principles (COMPSCI 230)

- Designed and led discussion sections
- Collaborated on the design of exam questions
- Held office hours for 50 students

Graduate Teaching Assistant 2016
Universidad de los Andes, Colombia
Course: Analysis and Processing of Biomedical Images (IBIO 3470)

- Designed and led discussion sections
- Designed and graded homework assignments
- Held office hours for 78 students

Invited Talks & Oral Presentations

AI-Guided Importance Sampling for Counting in Large Datasets

- Research Talk, RBC Borealis, Canada (Remote). Jan, 2026

From Data to Science with AI and Human-in-the-Loop

- Guest Lecture, Advanced Topics in Computer Vision, University of Michigan (CSE). January, 2026
- Invited Speaker, Biomedical Computer Vision Lab talk, Universidad de los Andes, Colombia. July, 2025
- Seminar Speaker, Masters in AI and Data Science, Universidad Autónoma de Occidente, Colombia. March, 2025
- Computer Vision Lab Seminar at the University of Michigan (CSE). July, 2024
- Computational MRI Lab Seminar at UC Berkeley (EECS). January, 2024

Motion Sensing and Correction using RF and Microwaves in MRI

- Invited Speaker, II Digital Health Colloquium, Universidad Autónoma de Occidente, Colombia. May, 2025

Introduction to Deep Learning and Applications in Biomedical Imaging

- Guest Speaker, Medical Imaging Analysis Using AI class, Universidad Autónoma de Occidente, Colombia. March, 2025

Seminar Speaker at Michigan Research and Discovery Scholar (MRADS) program. University of Michigan. February, 2024

DISCount : Counting in Large Image Collections with Detector-based Importance Sampling

- AAAI AI for Social Impact Track (AAAI-AISI). Vancouver, Canada. February 2024
- New England Computer Vision Workshop at Dartmouth College. Hanover, NH. December 2023
- UMass Machine Learning Club–Advanced Lecture. Amherst, MA. September 2023
- Probabilistic Machine Learning Lab Seminar at UMass Amherst. Amherst, MA. May 2023

Using Spatio-Temporal Information in Weather Radar Data to Detect Communal Bird Roosts

- 3rd International Radar Aeroecology Conference (IRAC 2022). Davos, Switzerland. (Remote). June 2022

Finding Four-Leaf Clovers : A Benchmark for Fine-Grained Object Localization

- Camera Trap Tech Symposium. Google Headquarters, Mountain View, CA, USA. (Remote). November 2019
- Fine-Grained Visual Categorization Workshop (FGVC6) at CVPR. Long Beach, CA. June 2019

Star Cluster Classification using Deep Learning

- Astronomy Department Meeting at UMass Amherst. Amherst, MA. June 2019

Automated detection of lung cancer using 3D ConvNets

- IEEE International Symposium on Biomedical Imaging (ISBI)–Lung Nodule Malignancy Prediction. Washington DC. April 2018
- 13th International Symposium on Medical Information Processing and Analysis. San Andres Island, Colombia. October 2017

Lung Nodules Detection in CT images using Computer Vision
- VIII International Seminar in Biomedical Engineering at Uniandes. Bogota, Colombia. April 2016

Grant Writing Experience

Contributed to proposal development, preliminary analyses, and technical sections for the following grants :
- NSF ACED, 2025. Grant archived
- NSF III Medium Grant, 2025. Funded, $1M (2025-2028)
- Samsung GRO, 2024. Not funded
- NSF Standard Grant, 2024. Funded, $421K (2024-2026)
- NSF Standard Grant, 2022. Not funded

Professional Experience (Pre-MSc/PhD)

Project Manager — 2010 – 2016
HORMESA America Ltd. Bogotá, Colombia

Junior Enginner — 2009 – 2010
HORMESA America Ltd. Bogotá, Colombia